

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        // Child Process
        printf("Child Before=%d\n", x); // Reads shared memory variable (x = 100)
        x = 500;                       // Triggers Copy-On-Write (COW) memory duplication
        printf("Child After=%d\n", x);  // Modifies child's private copy (x = 500)
    }
    else
    {
        // Parent Process
        sleep(2);                      // Sleep ensures child writes first
        printf("Parent=%d\n", x);      // Parent's variable remains unchanged (x = 100)
    }

    return 0;
}
```

```
[medikondarakesh@Medikondas-MacBook-Air ~ % nano cow_demo.c
[medikondarakesh@Medikondas-MacBook-Air ~ % gcc cow_demo.c -o cow_demo
[medikondarakesh@Medikondas-MacBook-Air ~ % ./cow_demo
Child Before=100
Child After=500
Parent=100
medikondarakesh@Medikondas-MacBook-Air ~ %
```